

Advay Kadam

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EDUCATION

University of Illinois Urbana-Champaign

Champaign, IL

B.S Computer Science and Statistics, Minor in Mathematics | GPA: 3.87

Expected Graduation: December 2026

- Relevant Courses: Algorithms and Models of Computation, Data Structures and Algorithms, Machine Learning, Abstract Linear Algebra, Computer Architecture, Time Series Analysis, Statistics and Probability

TECHNICAL SKILLS

Languages: Python, C++, SQL, R, Java, JavaScript, HTML/CSS

Tools: PyTorch, Tensorflow, NumPy, Pandas, Flask, Azure, AWS SageMaker, Apache Spark/PySpark, MySQL, Docker

EXPERIENCE

Bank of New York Mellon

June 2025 – Present

Data Science and Engineering Intern

Jersey City, NJ

- Developing pipeline integrating logistic regression with a RAG agent to automate metadata security classification across 4,000+ datasets, improving tagging accuracy by **10%** and reducing manual audit time by **15 hours/week**

Siebel Center of Computing and Data Science

Jan 2025 – Present

Course Assistant for Data Structures and Algorithms

Urbana, IL

- Led office hours and discussion sessions on abstract data types, C++ algorithms, and complexity analysis
- Created machine problems and auto-graded labs covering data structures such as graphs, trees, and hash maps
- Developed and maintained algorithmic exercises (e.g., nearest neighbor search) for **400+** enrolled students

Procter & Gamble

May 2024 – Aug 2024

Software Engineering Intern

Cincinnati, OH

- Increased sales ETL throughput **2.5x** on **200k+** monthly records by optimizing SQL transformations and implementing DAG-aware smart caching in PySpark; cut refresh time from **4 hours** to **90 minutes**
- Implemented statistical anomaly detection using Isolation Forests with Python to flag deviations in batch loads
- Documented **12-page** ETL architecture guide adopted across **3** teams to accelerate on-boarding and debugging
- Fine-tuned YOLOv9 model on retail image data to **95%** accuracy and executed model on Azure ML pipeline

National Center for Supercomputing Applications

Aug 2023 – May 2024

Undergraduate Researcher

Urbana, IL

- Applied statistical analysis to real-time network data to assess post-quantum cryptographic algorithm resilience
- Analyzed sample of **2+** petabytes of network metadata using Python, Pandas, NumPy, and Shell scripting
- Identified that over **99%** of SSH traffic lacked quantum-resistant encryption in collaboration with Dr. Phuong Cao
- Developed Flask dashboard to parse Zeek logs and deliver live metrics on PQC algorithm usage and vulnerability

Rice University

Nov 2021 – Mar 2023

Research Intern

Houston, TX

- Worked with Professor Aditya Mohite to enhance 2D-perovskite property predictions for photovoltaic applications
- Built PyTorch based regression model ($R^2 = 0.95$) to predict the energy-gap of 2D-halide perovskites
- Web-scraped and engineered **30+** chemical features using Pandas, NumPy, and Selenium for model training

PROJECTS AND LEADERSHIP

Disruption Lab at Gies | *Lead Software Engineer*

- Led a team of 6 students to train transfer learning model to detect AI-generated deepfake images with TensorFlow
- Trained and deployed model in AWS SageMaker, boosting accuracy to 96% via automated hyperparameter tuning

Regression Library | *C++, Xtensor, Boost, CMake*

- Built C++ library for multiple linear regression and hypothesis testing for reduced regression coefficients
- Implemented stochastic/batch gradient descent algorithms and confidence intervals for true parameter estimation

Time Series Regression Analysis of Sea Surface Temperatures | *R*

- Performed research on creating robust regression models for forecasting sea surface temperatures during El Niño
- Conducted residual diagnostics with ACF/PACF plots and Ljung-Box tests to validate autocorrelation structure

ACHIEVEMENTS

- **IMC Trading Prosperity Competition** | **Top 3%** for Algorithmic Trading Challenge
- **Datathons** | **3rd place** at UIUC Spring 2024 Datathon, **2nd place** at UIUC Fall 2023 Datathon
- **Hackathons** | Best AI Project Prize at PioneerHacks III, Best Environmental Project Prize at WildHacks II